STEFANO BUDI

stefano.budi@binus.ac.id | +62 822 2980 9271 | stefano-budi.vercel.app | github.com/avalon-py | linkedin.com/in/stefano-budi

EDUCATION

BINUS University — *Jakarta, Indonesia*
B.S. in Computer Science, Data Science Specialization Track — *2024 – 2028 (Expected)*
GPA: 3.99 / 4.00
Independent Study: MIT 15.401 Finance Theory I, Harvard STAT110 Probability, Stanford CS229 Machine Learning

PUBLICATIONS

"Dynamic Classifier Selection using Local Class Accuracy on Overlap Cleaned Heterogeneous Pool of Experts"
2nd Author | Published — *Sept 2026*

- Co-developed a dynamic classifier selection framework combining KNN-based overlap cleaning with local-class-accuracy-based expert selection across a heterogeneous pool of specialists (Naive Bayes, XGBoost, SVM), cutting log loss by 70.1% (0.898 → 0.269) versus a probability-harvesting baseline while achieving 91.62% accuracy and a macro F1-score of 0.8638 on the UCI Student Dropout dataset (n=4,424).

EXPERIENCE

Laboratory Assistant, BINUS University — *Feb 2025 – Feb 2026*

- Selected as 1 of 2 assistants from ~200 applicants (top <1%) through a 4-month, 3-stage process, including 1 month of full-time (7am–5pm) on-site bootcamp training, project creation, and presentation.
- Taught 5 classes weekly (~150 students) across Computer Vision and Algorithm & Programming (C), qualified in Deep Learning and Data Mining via BINUS's SLC qualifications; earned a 5.3/6 performance KPI.

PROJECTS

Speech2Market — End-to-End NLP-Based Market Impact Predictor for FED Speeches — *Aug 2026*
Dashboard | Live App | GitHub

- Engineered an NLP + time-series pipeline (classifier-less FinBERT embeddings, lagged feature engineering, macroeconomic indicators, HistGradientBoost) predicting SPX, VIX, Gold returns at 3/7/30-day horizons following The FED's speeches, sourced via a custom multi-era Fed website scraper with regex-based fallback parsing.
- Deployed under a 1GB memory constraint via FP16 ONNX quantization of FinBERT, and built an automated GitHub Actions pipeline to work around FRED API/Yahoo Finance API blacklist, running entirely on free-tier infrastructure.

FindIT — Face Anti-Spoofing Detection — *Apr 2026*
Computer vision competition — camera-based identity fraud detection (DAC Find IT! 2026)

- Built a 5-fold DINOv2-based face anti-spoofing classifier across 6 classes using a two-phase fine-tuning strategy — frozen-backbone head training followed by full fine-tuning with differential learning rates — plus weighted sampling and selective MixUp to address class imbalance.
- Achieved 0.9742 macro F1, placing 11th of 340 teams in the preliminary round (top 4%) — 0.005 behind 1st place, narrowly missing the top-10 cutoff for finals.

Laptop Price Predictor — End-to-End Ensemble ML Deployment — *May 2026*
Live App | GitHub

- Built a stacking ensemble (XGBoost, LightGBM, CatBoost → RidgeCV meta-learner) tuned via Optuna Bayesian optimization, achieving 6.67% MAPE and outperforming all base models in accuracy and cross-validation stability.
- Shipped a full pipeline from training with asynchronous web-scraped dataset (BeautifulSoup + Playwright, 583 listings) to a deployed Streamlit app (P90 latency ~65–75ms), validated via a 5-user study with 100% "useful" rating.

Makan Bergizi Gratis (MBG) Investment Priority Mapping — Graph-Based Clustering Research — *Jul 2025*
Dashboard | GitHub

- Applied spectral (graph-based) clustering with a Laplacian structure to map investment priority across 38 Indonesian provinces, engineering 15+ derived features from raw government data (BPS, Badan Pangan Nasional), filtered via RFECV, validated via the Calinski-Harabasz index (3.94 vs. a 1.01 random-baseline benchmark).
- Selected by BINUS's Computer Science faculty as product presentation material in Pameran Produk Inovasi (PPI) by Badan Riset dan Inovasi Daerah (BRIDA), presenting to government stakeholders and civil servants.

TECHNICAL SKILLS

Languages: Python, SQL (fluent); C (proficient); Java (basic); RapidMiner, R (familiar)
Machine Learning: Gradient Boosting, Spectral Clustering, Temporal Fusion Transformers, Naïve Bayes, Decision Trees, SVM, PCA/LDA, Optuna
Transfer Learning: DINOv2, ConvNeXt, FinBERT, ONNX Runtime, LLM Application Integration
Data Engineering & Deployment: Pandas, BeautifulSoup, Playwright, GitHub Actions, Streamlit, Git/GitHub